

Quantified Linear Arithmetic

Whilst in [Presburger arithmetic] one can only state rather simple theorems, yet an efficient algorithm... is useful in that it can quickly dispose of a host of the simpler formulas..., leaving only the more complex to be dealt with by some more general theorem prover or by the human.

— David C. Cooper

Theorem Proving in Arithmetic Without Multiplication, 1972

Recall from Chapter 3 that satisfiability in the theories of linear integer arithmetic $T_{\mathbb{Z}}$ and linear rational (or real) arithmetic $T_{\mathbb{Q}}$ is decidable. This chapter explores a common method of deciding satisfiability of arbitrarily quantified formulae in decidable theories in the context of $T_{\mathbb{Z}}$ and $T_{\mathbb{Q}}$.

A **quantifier elimination procedure** is an algorithm that constructs from a given formula an equivalent quantifier-free formula. When the given formula does not have any free variables, the atoms of the resulting formula are applications of predicates to constant terms such as $3 < 5$. If the truth value of these constant terms is decidable, a quantifier elimination procedure provides a basis for a satisfiability decidable procedure. When the given formula contains free variables, the resulting quantifier-free formula contains a subset of these free variables.

Section 7.1 presents the method of quantifier elimination in an abstract context. Sections 7.2 and 7.3 then present quantifier elimination methods for the theory of integers $T_{\mathbb{Z}}$ and the theory of rationals $T_{\mathbb{Q}}$, respectively. The complexity results of Section 7.4 suggest that these algorithms are near-optimal in time complexity, although quantifier-elimination methods are, in general, computationally expensive.

Sections 7.2.4 and 7.2.5 discuss optimizations for the integer case that are particularly effective when the given formula has only existential quantifiers or only universal quantifiers. With these optimizations, the quantifier elimination method for $T_{\mathbb{Z}}$ is our main decision procedure for the quantifier-free fragment

of $T_{\mathbb{Z}}$. However, the case for $T_{\mathbb{Q}}$ is different: Chapter 8 presents a special decision procedure for the quantifier-free fragment of $T_{\mathbb{Q}}$.

7.1 Quantifier Elimination

7.1.1 Quantifier Elimination

Quantifier elimination (QE) is the main technique that underlies the algorithms of this chapter. As the name suggests, the idea is to eliminate quantifiers of a formula F until only a quantifier-free formula G that is equivalent to F remains.

Formally, a theory T **admits quantifier elimination** if there is an algorithm that, given Σ -formula F , returns a quantifier-free Σ -formula G that is T -equivalent to F . Then T is decidable if satisfiability in the quantifier-free fragment of T is decidable.

Example 7.1. Consider the $\Sigma_{\mathbb{Q}}$ -formula

$$F : \exists x. 2x = y ,$$

which expresses the set of rationals y that can be halved. Intuitively, all rationals can be halved, so a quantifier-free $T_{\mathbb{Q}}$ -equivalent formula is

$$G : \top ,$$

which expresses the set of all rationals. Also, G states that F is valid. ■

Example 7.2. Consider the $\Sigma_{\mathbb{Z}}$ -formula

$$F : \exists x. 2x = y ,$$

which expresses the set of integers y that can be halved (to produce another integer). Intuitively, only even integers can be halved. Can you think of a quantifier-free $T_{\mathbb{Z}}$ -equivalent formula to F ? In fact, no such formula exists. Later, we introduce an augmented theory of integers that contains a countably infinite number of divisibility predicates. For example, in this extended theory, an equivalent formula to F is

$$G : 2 \mid y ,$$

which expresses the set of even integers: integers that are divisible by 2. ■

7.1.2 A Simplification

In developing a QE algorithm for theory T , we need only consider formulae of the form $\exists x. F$ for quantifier-free formula F . For given arbitrary formula G , choose the innermost quantified formula $\exists x. H$ or $\forall x. H$. In the latter case, rewrite $\forall x. H$ as $\neg(\exists x. \neg H)$ and focus on the subformula $\exists x. \neg H$ inside the negation.

In both the existential and universal cases, we now have a formula of the form $\exists x. F$ to which we apply the QE algorithm to find T -equivalent and quantifier-free formula H' . In the existential case, replace $\exists x. H$ in G with H' . In the universal case, replace $\forall x. H$ in G with $\neg H'$.

Example 7.3. Consider the Σ -formula

$$G_1 : \exists x. \forall y. \exists z. F_1[x, y, z] ,$$

and assume that T admits quantifier elimination. The innermost quantified formula is $\exists z. F_1[x, y, z]$. Applying the QE algorithm for T to this subformula returns $F_2[x, y]$:

$$G_2 : \exists x. \forall y. F_2[x, y] .$$

The innermost quantified formula is now $\forall y. F_2[x, y]$; rewriting, we have

$$G_3 : \exists x. \neg(\exists y. \neg F_2[x, y]) .$$

Applying the QE algorithm to the existential subformula $\exists y. \neg F_2[x, y]$ produces $F_3[x]$. We now have

$$G_4 : \exists x. \neg F_3[x] .$$

Finally, applying the QE algorithm one more time to G_4 produces a quantifier-free formula G_5 . G_5 is T -equivalent to G_1 . ■

7.2 Quantifier Elimination over Integers

We now turn to the theory of integers $T_{\mathbb{Z}}$. Presburger showed this theory to be decidable in 1929. We discuss the QE procedure of Cooper, first described in 1971.

7.2.1 Augmented Theory of Integers

Recall that $T_{\mathbb{Z}}$ has the following signature:

$$\Sigma_{\mathbb{Z}} : \{ \dots, -2, -1, 0, 1, 2, \dots, -3, -2, 2, 3, \dots, +, -, =, < \} ,$$

where

- $\dots, -2, -1, 0, 1, 2, \dots$ are constants, intended to be assigned the obvious corresponding values in the intended domain $\mathbb{Z}$;
- $\dots, -3\cdot, -2\cdot, 2\cdot, 3\cdot, \dots$ are unary functions, intended to represent constant coefficients (e.g., $2 \cdot x$, abbreviated $2x$);
- $+$ and $-$ are binary functions, intended to represent the obvious corresponding functions over $\mathbb{Z}$;
- $=$ and $>$ are binary predicates, intended to represent the obvious corresponding predicates over $\mathbb{Z}$.

Example 7.2 claims that the $\Sigma_{\mathbb{Z}}$ -formula $\exists x. 2x = y$ does not have a $T_{\mathbb{Z}}$ -equivalent quantifier-free formula. The following lemma proves a more general claim.

Lemma 7.4. *Consider quantifier-free $\Sigma_{\mathbb{Z}}$ -formula F such that $\text{free}(F) = \{y\}$. F represents the set of integers*

$$S : \{n \in \mathbb{Z} : F\{y \mapsto n\} \text{ is } T_{\mathbb{Z}}\text{-valid}\} .$$

Either $S \cap \mathbb{Z}^+$ or $\mathbb{Z}^+ \setminus S$ is finite.

$\mathbb{Z}^+$ is the set of positive integers; $\cap$ and $\setminus$ are set intersection and complement, respectively. The lemma says that every quantifier-free $\Sigma_{\mathbb{Z}}$ -formula with only one free variable represents a set of integers S such that either the subset of positive integers in S has finite cardinality or the set of positive integers not in S has finite cardinality. Exercise 7.1 asks the reader to prove this lemma.

Consider again the case of $\exists x. 2x = y$, and let S be the set of integers satisfying the formula, namely the even integers. Since both the set of positive even integers $S \cap \mathbb{Z}^+$ and the set of positive odd integers $\mathbb{Z}^+ \setminus S$ are infinite, the set of even integers cannot be represented in $T_{\mathbb{Z}}$ by a quantifier-free formula according to the lemma. Therefore, there is no quantifier-free $\Sigma_{\mathbb{Z}}$ -formula that is $T_{\mathbb{Z}}$ -equivalent to $\exists x. 2x = y$, and thus $T_{\mathbb{Z}}$ does not admit QE.

To circumvent this problem, we augment the theory $T_{\mathbb{Z}}$ with an infinite but countable number of unary **divisibility predicates**

$$k \mid \cdot \quad \text{for } k \in \mathbb{Z}^+ ;$$

that is, a predicate exists for each positive integer k . The intended interpretation of $k \mid x$ is that it holds iff k divides x without any remainder. For example,

$$x > 1 \wedge y > 1 \wedge 2 \mid x + y$$

is satisfiable (choose $x = 2$ and $y = 2$), but

$$\neg(2 \mid x) \wedge 4 \mid x$$

is not satisfiable.

The augmented theory $\widehat{T}_{\mathbb{Z}}$ extends the signature of $T_{\mathbb{Z}}$ with these divisibility predicates. Additionally, the predicates are defined by the countable set of axioms

$$\forall x. k \mid x \leftrightarrow \exists y. x = ky \quad (\text{divides})$$

for $k \in \mathbb{Z}^+$. The next section shows that $\widehat{T}_{\mathbb{Z}}$ admits QE.

7.2.2 Cooper's Method

In this section, we present the basic form of Cooper's quantifier elimination method. In the subsequent sections, we examine a set of optimizations.

The algorithm is given a $\widehat{\Sigma}_{\mathbb{Z}}$ -formula $\exists x. F[x]$ as input, where F is quantifier-free but may contain free variables in addition to x . It then proceeds to construct a quantifier-free $\widehat{\Sigma}_{\mathbb{Z}}$ -formula that is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$ according to the following steps.

Step 1

Put $F[x]$ in NNF. The output $\exists x. F_1[x]$ is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$ and is such that F_1 is a positive Boolean combination (only $\wedge$ and $\vee$) of literals.

Step 2

Replace literals according to the following $\widehat{T}_{\mathbb{Z}}$ -equivalences, applied from left to right:

$$\begin{aligned} s = t &\Leftrightarrow s < t + 1 \wedge t < s + 1 \\ \neg(s = t) &\Leftrightarrow s < t \vee t < s \\ \neg(s < t) &\Leftrightarrow t < s + 1 \end{aligned}$$

The output $\exists x. F_2[x]$ is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$ and contains only literals of the form

$$s < t, \quad k \mid t, \quad \text{or} \quad \neg(k \mid t),$$

where s, t are $\widehat{\Sigma}_{\mathbb{Z}}$ -terms and $k \in \mathbb{Z}^+$.

Example 7.5. Applying the $\widehat{T}_{\mathbb{Z}}$ -equivalences to

$$\neg(x < y) \wedge \neg(x = y + 3)$$

produces the $\widehat{T}_{\mathbb{Z}}$ -equivalent formula

$$y < x + 1 \wedge (x < y + 3 \vee y + 3 < x).$$

■

Step 3

Collect terms containing x so that literals have the form

$$hx < t, \quad t < hx, \quad k \mid hx + t, \quad \text{or} \quad \neg(k \mid hx + t),$$

where t is a term that does not contain x and $h, k \in \mathbb{Z}^+$. The output is the formula $\exists x. F_3[x]$, which is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$.

Example 7.6. Collecting terms in

$$x + x + y < z + 3z + 2y - 4x$$

produces the $\widehat{T}_{\mathbb{Z}}$ -equivalent formula

$$6x < 4z + y.$$

■

Step 4

Let

$$\delta' = \text{lcm}\{h : h \text{ is a coefficient of } x \text{ in } F_3[x]\},$$

where lcm returns the least common multiple of the set. Multiply atoms in $F_3[x]$ by constants so that δ' is the coefficient of x everywhere:

$$\begin{array}{lll} hx < t & \Leftrightarrow & \delta'x < h't & \text{where } h'h = \delta' \\ t < hx & \Leftrightarrow & h't < \delta'x & \text{where } h'h = \delta' \\ k \mid hx + t & \Leftrightarrow & h'k \mid \delta'x + h't & \text{where } h'h = \delta' \\ \neg(k \mid hx + t) & \Leftrightarrow & \neg(h'k \mid \delta'x + h't) & \text{where } h'h = \delta' \end{array}$$

Notice the abuse of notation: $h'k \mid \cdot$ is a different (unary) predicate than $k \mid \cdot$. This rewriting results in formula F'_3 in which all occurrences of x occur in terms $\delta'x$. Replace $\delta'x$ terms with a fresh variable x' to form F''_3 :

$$F''_3 : F'_3\{\delta'x \mapsto x'\}.$$

Finally, construct

$$\exists x'. \underbrace{F''_3[x'] \wedge \delta' \mid x'}_{F_4[x']}.$$

The divisibility literal constrains the fresh variable x' to be divisible by δ' , which exactly captures the values of $\delta'x$.

$\exists x'. F_4[x']$ is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$. Moreover, each literal of $F_4[x']$ that contains x' has one of the following forms:

- (A) $x' < a$
- (B) $b < x'$
- (C) $h \mid x' + c$
- (D) $\neg(k \mid x' + d)$

where a, b, c, d are terms that do not contain x , and $h, k \in \mathbb{Z}^+$.

Step 5

Construct the **left infinite projection** $F_{-\infty}[x']$ from $F_4[x']$ by replacing

(**A**) literals $x' < a$ by $\top$

and

(**B**) literals $b < x'$ by $\perp$.

The idea is that very small numbers (the left side of the “number line”) satisfy (**A**) literals but not (**B**) literals.

Let

$$\delta = \text{lcm} \left\{ \begin{array}{l} h \text{ of } (\mathbf{C}) \text{ literals } h \mid x' + c \\ k \text{ of } (\mathbf{D}) \text{ literals } \neg(k \mid x' + d) \end{array} \right\}$$

and B be the set of b terms appearing in (**B**) literals. Construct

$$F_5 : \bigvee_{j=1}^{\delta} F_{-\infty}[j] \vee \bigvee_{j=1}^{\delta} \bigvee_{b \in B} F_4[b + j] .$$

F_5 is quantifier-free and $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$.

Step 5 is the trickiest part of the procedure, so let us focus on this step. The first major disjunct of F_5 contains only divisibility literals. It asserts that an infinite number of small numbers n satisfy $F_4[n]$. For if there exists one number n that satisfies the Boolean combination of divisibility literals in $F_{-\infty}$, then every $n - \lambda\delta$, for $\lambda \in \mathbb{Z}^+$, also satisfies $F_{-\infty}$.

The second major disjunct asserts that there is a least $n \in \mathbb{Z}$ that satisfies $F_4[n]$. This least n is determined by the b terms of the (**B**) literals.

More formally, consider the following **periodicity property** of the divisibility predicates:

If $m \mid \delta$, then $m \mid n$ iff $m \mid n + \lambda\delta$ for all $\lambda \in \mathbb{Z}$.

In other words, $m \mid \cdot$ cannot distinguish between the cases $m \mid n$ and $m \mid n + \lambda\delta$. Since δ is chosen in Step 5 to be the least common multiple of the integers h and k of the divides predicates, no divides literal in F_5 can distinguish between two integers n and $n + \lambda\delta$, for any $\lambda \in \mathbb{Z}$.

With this property in mind, consider the first major disjunct of F_5 again. If $n \in \mathbb{Z}$ satisfies $F[n]$, then so does $n - \lambda\delta$ for $\lambda \in \mathbb{Z}^+$. Then surely a small enough number exists that satisfies all (**A**) literals and falsifies all (**B**) literals of F_4 , mirroring the construction of $F_{-\infty}$.

For the second major disjunct, suppose that some number n satisfies $F_4[n]$. Decreasing this number continues to satisfy the same (**A**) literals. It cannot decrease past some value b^* without changing the truth of some (**B**) literal.

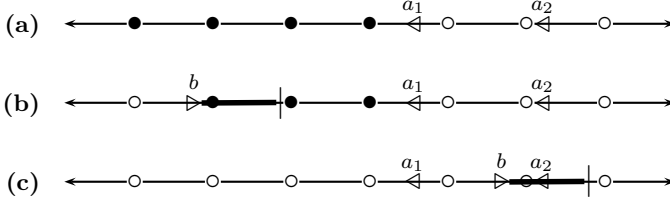


Fig. 7.1. (a) Left infinite projection (b) δ -interval (c) false

However, according to the periodicity rule, there exists a satisfying integer n' within the δ -interval to the right of b^* .

Figure 7.1 illustrates several situations when a , b , c , and d terms are constant. Circles represent points that satisfy the divides constraints; solid circles in particular represent satisfying points. Figure 7.1(a) illustrates a formula $x < a_1 \wedge x < a_2 \wedge \delta \mid x$: each left-pointing triangle represents a $x < a_i$ literal. The left infinite projection is satisfied. Figure 7.1(b) illustrates an additional $x > b$ literal; now, the δ -interval following the right-pointing triangle at b is searched. It contains a satisfying point. Finally, $b > a_1$ in Figure 7.1(c), so the δ -interval does not contain a satisfying point.

Example 7.7. Consider $\widehat{\Sigma_{\mathbb{Z}}}$ -formula

$$\exists x. \underbrace{3x - 2y + 1 > -y \wedge 2x - 6 < z \wedge 4 \mid 5x + 1}_{F[x]} .$$

After Step 3, we have

$$\exists x. \underbrace{2x < z + 6 \wedge y - 1 < 3x \wedge 4 \mid 5x + 1}_{F_3[x]} .$$

Collecting coefficients of x in Step 4, we find

$$\delta' = \text{lcm}\{2, 3, 5\} = 30 .$$

Multiplying when necessary, we rewrite the formula so that 30 is the coefficient of every occurrence of x :

$$\exists x. 30x < 15z + 90 \wedge 10y - 10 < 30x \wedge 24 \mid 30x + 6 .$$

Replacing $30x$ with fresh x' and conjoining a divides atom completes Step 4:

$$\exists x'. \underbrace{x' < 15z + 90 \wedge 10y - 10 < x' \wedge 24 \mid x' + 6 \wedge 30 \mid x'}_{F_4[x']} .$$

For Step 5, construct the left infinite projection

$$F_{-\infty}[x] : \top \wedge \perp \wedge 24 \mid x' + 6 \wedge 30 \mid x' ,$$

which simplifies to $\perp$. Compute

$$\delta = \text{lcm}\{24, 30\} = 120$$

and

$$B = \{10y - 10\} .$$

Then replacing x' by $10y - 10 + j$ in $F_4[x']$ produces

$$F_5 : \bigvee_{j=1}^{120} \left[\begin{array}{l} 10y - 10 + j < 15z + 90 \wedge 10y - 10 < 10y - 10 + j \\ \wedge 24 \mid 10y - 10 + j + 6 \wedge 30 \mid 10y - 10 + j \end{array} \right]$$

which simplifies to

$$F_5 : \bigvee_{j=1}^{120} \left[\begin{array}{l} 10y + j < 15z + 100 \wedge 0 < j \\ \wedge 24 \mid 10y + j - 4 \wedge 30 \mid 10y + j - 10 \end{array} \right] .$$

F_5 is quantifier-free and $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$. ■

Example 7.8. Consider again the formula defining the set of even integers:

$$\exists x. \underbrace{2x = y}_{F[x]} .$$

Rewriting according to Steps 2 and 3 produces

$$\exists x. y - 1 < 2x \wedge 2x < y + 1 .$$

Then

$$\delta' = \text{lcm}\{2, 2\} = 2 ,$$

so Step 4 completes with

$$\exists x'. \underbrace{y - 1 < x' \wedge x' < y + 1 \wedge 2 \mid x'}_{F_4[x']} .$$

Computing the left infinite projection $F_{-\infty}$ produces $\perp$, as $F_4[x']$ contains a **(B)** literal as a conjunct. However,

$$\delta = \text{lcm}\{2\} = 2$$

and

$$B = \{y - 1\} ,$$

so

$$F_5 : \bigvee_{j=1}^2 (y-1 < y-1+j \wedge y-1+j < y+1 \wedge 2 \mid y-1+j) .$$

Simplifying, we find

$$F_5 : \bigvee_{j=1}^2 (0 < j \wedge j < 2 \wedge 2 \mid y+j-1) ,$$

and then

$$F_5 : 2 \mid y ,$$

which is quantifier-free and $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists x. F[x]$. ■

Example 7.9. Consider the formula

$$\exists x. \underbrace{(3x+1 < 10 \vee 7x-6 > 7) \wedge 2 \mid x}_{F[x]} .$$

Rewriting to isolate x terms produces

$$\exists x. (3x < 9 \vee 13 < 7x) \wedge 2 \mid x ,$$

so

$$\delta' = \text{lcm}\{3, 7\} = 21 .$$

After multiplying coefficients by proper constants,

$$\exists x. (21x < 63 \vee 39 < 21x) \wedge 42 \mid 21x ,$$

we replace $21x$ by x' :

$$\exists x'. \underbrace{(x' < 63 \vee 39 < x') \wedge 42 \mid x' \wedge 21 \mid x'}_{F_4[x']} .$$

Then

$$F_{-\infty}[x'] : (\top \vee \perp) \wedge 42 \mid x' \wedge 21 \mid x' ,$$

or, simplifying,

$$F_{-\infty}[x'] : 42 \mid x' \wedge 21 \mid x' .$$

Finally,

$$\delta = \text{lcm}\{21, 42\} = 42$$

and

$$B = \{39\} ,$$

so

$$F_5 : \bigvee_{j=1}^{42} (42 \mid j \wedge 21 \mid j) \vee \bigvee_{j=1}^{42} ((39 + j < 63 \vee 39 < 39 + j) \wedge 42 \mid 39 + j \wedge 21 \mid 39 + j) .$$

Since $42 \mid 42$ and $21 \mid 42$, the left main disjunct simplifies to $\top$, so that $\exists x. F[x]$ is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\top$. Thus, F is $\widehat{T}_{\mathbb{Z}}$ -valid. $\blacksquare$

Theorem 7.10 (Correct). *Given $\widehat{\Sigma}_{\mathbb{Z}}$ -formula $\exists x. F[x]$ in which F is quantifier-free, Cooper's method returns a $\widehat{T}_{\mathbb{Z}}$ -equivalent quantifier-free formula.*

Proof. The transformations of the first four steps produce formula F_4 . By inspection, we assert that in $\widehat{T}_{\mathbb{Z}}$

$$\exists x. F[x] \Leftrightarrow \exists x. F_4[x] .$$

The focus of the proof is to prove that $\exists x. F_4[x] \Leftrightarrow F_5$ in $\widehat{T}_{\mathbb{Z}}$:

$$\exists x. F_4[x] \Leftrightarrow \bigvee_{j=1}^{\delta} F_{-\infty}[j] \vee \bigvee_{j=1}^{\delta} \bigvee_{b \in B} F_4[b + j] .$$

We accomplish the proof in two steps.

1. $F_5 \Rightarrow \exists x. F_4[x]$: We assume the existence of an interpretation I such that $I \models F_5$ and prove that $I \models \exists x. F_4[x]$.
2. $\exists x. F_4[x] \Rightarrow F_5$: We assume the existence of an interpretation I such that $I \models \exists x. F_4[x]$ and prove that $I \models F_5$.

Assume then that $I \models F_5$, so that one of the disjuncts of F_5 is **true** under I . If one of the second set of disjuncts is **true**, say $F_4[b^* + j^*]$, then

$$\begin{aligned} I \triangleleft \{x \mapsto b^* + j^*\} &\models F_4[x] \\ I &\models \exists x. F_4[x] . \end{aligned}$$

Otherwise, one of the first set of disjuncts is **true**, so for some $j^* \in [1, \delta]$, $I \triangleleft \{x \mapsto j^*\} \models F_{-\infty}[x]$. By construction of $F_{-\infty}$, there is some $\lambda > 0$ such that $I \triangleleft \{x \mapsto j^* - \lambda\delta\} \models F_4[x]$. That is, there is some $j^* - \lambda\delta$ that is so small that the inequality literals of F_4 evaluate under $I \triangleleft \{x \mapsto j^* - \lambda\delta\}$ exactly as in the construction of $F_{-\infty}$. Thus, $I \models \exists x. F_4[x]$ in this case as well.

For the other direction, assume that $I \models \exists x. F_4[x]$. Thus, some $n \in \mathbb{Z}$ exists such that $I \triangleleft \{x \mapsto n\} \models F_4[x]$. If for some $b^* \in B$ and $j^* \in [1, \delta]$,

$I \models n = b^* + j^*$, then $I \models F_4[b^* + j^*]$. As $F_4[b^* + j^*]$ is a disjunct of F_5 , $I \models F_5$.

Otherwise, consider whether $I \triangleleft \{x \mapsto n - \delta\} \models F_4[x]$. If not, then one of the **(B)** literals, say $b^* < x$ for some $b^* \in B$, of F_4 becomes false under I in the transition from n to $n - \delta$. But then $I \models n = b^* + j^*$ for some $j^* \in [1, \delta]$, contradicting our assumption that n is not equal to some $b^* + j^*$. Hence, it must be the case that $I \triangleleft \{x \mapsto n - \delta\} \models F_4[x]$.

By induction using this argument, we find that $I \triangleleft \{x \mapsto n - \lambda\delta\} \models F_4[x]$ for all $\lambda > 0$. For some λ , $n - \lambda\delta$ becomes so small that

$$I \triangleleft \{x \mapsto n - \lambda\delta\} \models F_4[x] \leftrightarrow F_{-\infty}[x] , \text{ so}$$

$$I \triangleleft \{x \mapsto n - \lambda\delta\} \models F_{-\infty}[x] .$$

That is, $n - \lambda\delta$ is so small that the inequality literals of F_4 evaluate under $I \triangleleft \{x \mapsto n - \lambda\delta\}$ exactly as in the construction of $F_{-\infty}$. Now, since $F_{-\infty}$ contains only divides literals, we can choose a μ such that $n - \lambda\delta + \mu\delta \in [1, \delta]$. Let $j^* = n - \lambda\delta + \mu\delta$. Then $I \models F_{-\infty}[j^*]$, so that $I \models F_5$. ■

7.2.3 A Symmetric Elimination

The construction in Step 5 was biased to the left. We can just as easily define a right elimination. Construct the **right infinite projection** $F_{+\infty}[x']$ from $F_4[x']$ by replacing

(A) literals $x' < a$ by $\perp$

and

(B) literals $b < x'$ by $\top$.

The idea is that very large numbers (the right side of the “number line”) satisfy **(B)** literals but not **(A)** literals.

Then define δ as before, but now define A as the set of a terms appearing in **(A)** literals. Construct

$$F_5 : \bigvee_{j=1}^{\delta} F_{+\infty}[-j] \vee \bigvee_{j=1}^{\delta} \bigvee_{a \in A} F_4[a - j] .$$

Now, instead of choosing the **left** or **right elimination** (corresponding to the left or right infinite projections, respectively) arbitrarily, choose the elimination according to the number of **(A)** and **(B)** literals. If there are fewer **(A)** literals than **(B)** literals, choose the right elimination; otherwise, choose the left elimination. This heuristic minimizes the number of disjuncts in the resulting formula F_5 .

Example 7.11. In the formula

$$F : \exists x. (x < 13 \vee x > 15) \wedge x < y ,$$

there are two **(A)** literals but only one **(B)** literal. Hence, choose the left infinite projection to produce fewer disjuncts. ■

7.2.4 Eliminating Blocks of Quantifiers

Consider a formula with a block of quantifiers: $\exists x_1. \dots \exists x_n. F[x_1, \dots, x_n]$, where F is quantifier-free. Eliminating x_n produces a formula of the form

$$G_1 : \exists x_1. \dots \exists x_{n-1}. \bigvee_{j=1}^{\delta} F_{-\infty}[x_1, \dots, x_{n-1}, j] \vee \bigvee_{j=1}^{\delta} \bigvee_{b \in B} F_4[x_1, \dots, x_{n-1}, b + j] .$$

Disjunction and existential quantification commute, so we can rewrite G_1 as

$$G_2 : \bigvee_{j=1}^{\delta} \exists x_1. \dots \exists x_{n-1}. F_{-\infty}[x_1, \dots, x_{n-1}, j] \vee \bigvee_{j=1}^{\delta} \bigvee_{b \in B} \exists x_1. \dots \exists x_{n-1}. F_4[x_1, \dots, x_{n-1}, b + j]$$

and continue the elimination on each disjunct.

This optimization can be taken one step further. Rather than expanding the disjuncts over the iterator j , treat j as a free variable during the subsequent eliminations. Then only $1 + |B|$ formulae need be examined during the next phase: the formula $\exists x_1. \dots \exists x_{n-1}. F_{-\infty}[x_1, \dots, x_{n-1}, j]$ and, for each $b \in B$, the formula $\exists x_1. \dots \exists x_{n-1}. F_4[x_1, \dots, x_{n-1}, b + j]$.

Example 7.12. Consider the formula

$$\exists y. \exists x. \underbrace{x < -2 \wedge 1 - 5y < x \wedge 1 + y < 13x}_{F[x,y]} .$$

At Step 3,

$$\delta' = \text{lcm}\{1, 13\} = 13 ,$$

producing

$$\exists y. \exists x. 13x < -26 \wedge 13 - 65y < 13x \wedge 1 + y < 13x$$

and then

$$\exists y. \exists x'. x' < -26 \wedge 13 - 65y < x' \wedge 1 + y < x' \wedge 13 \mid x' .$$

With $\delta = \text{lcm}\{13\} = 13$, $A = \{-26\}$, and $B = \{13 - 65y, 1 + y\}$, choose the right elimination to form:

$$\exists y. \bigvee_{j=1}^{13} \left[\begin{array}{l} -26 - j < -26 \wedge 13 - 65y < -26 - j \\ \wedge 1 + y < -26 - j \wedge 13 \mid -26 - j \end{array} \right] .$$

$F_{+\infty}$ simplifies to $\perp$ since F is a conjunction of both **(A)** and **(B)** literals. Now, instead of applying elimination to the entire subformula within the quantifier $\exists y$, commute the quantifier and the disjunctions:

$$G : \bigvee_{j=1}^{13} \exists y. j > 0 \wedge 39 + j < 65y \wedge y < -27 - j \wedge 13 \mid -j - 26 .$$

Treating j as a free variable, apply QE to the subformula

$$H : \exists y. j > 0 \wedge 39 + j < 65y \wedge y < -27 - j \wedge 13 \mid -j - 26$$

as usual. Then simplify to produce

$$H' : \bigvee_{k=1}^{65} (k < -66j - 1794 \wedge 13 \mid -j - 26 \wedge 65 \mid 39 + j + k) ,$$

and replace H with H' in G to produce the final formula

$$\bigvee_{j=1}^{13} \bigvee_{k=1}^{65} (k < -66j - 1794 \wedge 13 \mid -j - 26 \wedge 65 \mid 39 + j + k) .$$

This formula is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\exists y, x. F[x, y]$. ■

7.2.5 ★Solving Divides Constraints

Consider a formula of the form $G : \exists x_1. \dots \exists x_n. F[x_1, \dots, x_n]$ without free variables. Applying Cooper's method with the block elimination optimization produces a quantifier-free $\widehat{T}_{\mathbb{Z}}$ -equivalent formula of the form

$$G' : \bigvee_{j_1=1}^{\delta_1} \dots \bigvee_{j_n=1}^{\delta_n} F'[j_1, \dots, j_n] ,$$

also without free variables. Expanding this formula by attempting every possible combination of values for $j_1, \dots, j_n$ produces $\delta_1 \times \delta_2 \times \dots \times \delta_n$ disjuncts. This naive expansion is prohibitively expensive on even small problems.

Notice, however, that Step 4 introduces many divisibility literals as conjuncts. F_5 has the form

$$F_5 : \bigvee_{j_1=1}^{\delta_1} \cdots \bigvee_{j_n=1}^{\delta_n} \left(F'' \wedge \bigwedge_i k_i \mid t_i[j_1, \dots, j_n] \right),$$

where the t_i are terms containing only constants and the j_i iterators. Cooper realized that the conjuncts

$$D : \bigwedge_i k_i \mid t_i[j_1, \dots, j_n]$$

can be solved to reduce significantly the number of disjuncts to consider.

There are two steps to solving divides constraints.

Step 1: Triangulate the Constraints

The following theorem provides a means for reducing the number of literals that contain some j_i . It applies Euclid's algorithm for computing the greatest common divisor (GCD) d of two integers m and n . Euclid's algorithm also returns two integers p and q such that $pm + qn = d$.

Theorem 7.13. *Consider two divisibility constraints*

$$F : m \mid ax + b \wedge n \mid \alpha x + \beta,$$

where $m, n \in \mathbb{Z}^+$, $a, \alpha \in \mathbb{Z} \setminus \{0\}$, and b, β are terms not containing x . Let $d, p, q = \text{gcd}(an, \alpha m)$ be such that d is the GCD of an and αm , and p and q obey $pan + q\alpha m = d$. Then F is satisfiable iff

$$G : mn \mid dx + bpn + \beta qm \wedge d \mid \alpha b - a\beta$$

is satisfiable.

While both of the literals of F contain x , only one of the literals of G contains x . Therefore, we can apply this theorem to triangulate a set S of divisibility constraints. Let $\prec$ be a linear ordering of $j_1, \dots, j_n$. S is in **triangular form** if for each j_i , at most one constraint of S contains j_i as the least (according to $\prec$) free variable.

The triangularization algorithm proceeds iteratively. On one iteration, perform the following steps:

1. Choose from S two constraints

$$m \mid aj_i + b \quad \text{and} \quad n \mid \alpha j_i + \beta$$

such that there is no $j_k \prec j_i$ that occurs in at least two divisibility constraints of S .

2. Apply Theorem 7.13 to produce the new constraints

$$mn \mid dj_i + bpn + \beta qm \quad \text{and} \quad d \mid \alpha b - a\beta.$$

Replace the original constraints with these constraints in S .

Example 7.14. Consider the divisibility constraints of Example 7.12:

$$13 \mid -j - 26 \wedge 65 \mid 39 + j + k .$$

Fix the variable order $j \prec k$. According to this order, only one constraint should have an occurrence of j , so apply Theorem 7.13:

$$\begin{array}{l} m \mid aj + b \quad n \mid \alpha j + \beta \\ 13 \mid -j + -26 \wedge 65 \mid j + k + 39 \end{array}$$

Compute

$$\begin{array}{l} d, p, q = \gcd(an, \alpha m) \\ 13, 0, 1 = \gcd(-65, 13) \end{array} \quad \text{so that} \quad \begin{array}{l} pan + q\alpha m = d \\ 0(-65) + 1(13) = 13 \end{array}$$

and construct

$$\begin{array}{l} mn \mid dj + bpn + \beta qm \quad d \mid \alpha b - a\beta \\ 845 \mid 13j + (-26)(0)(65) + (k + 39)13 \wedge 13 \mid -26 - (-1)(k + 39) \end{array}$$

or, simplifying,

$$845 \mid 13j + 13k + 507 \wedge 13 \mid k + 13 .$$

As desired, only one constraint has an occurrence of j . Additionally, the simplest constraint contains only k . ■

Step 2: Solve the Constraints

To solve the triangulated system of constraints S , consider the following theorem.

Theorem 7.15. *Consider divisibility constraint*

$$F : m \mid ax + b ,$$

for $m \in \mathbb{Z}^+$, $a \in \mathbb{Z} \setminus \{0\}$, and b a term not containing x . Let $d, p, q = \gcd(a, m)$ be such that d is the GCD of a and m , and p and q obey $pa + qm = d$. F is satisfiable iff $d \mid b$. If it is satisfiable, its solutions are

$$x = -\frac{pb}{d} + \frac{\lambda m}{d} \quad \text{for } \lambda \in \mathbb{Z} .$$

With S in triangular form, we need only solve the system recursively:

1. If S is empty, generate a disjunct according to the current values of $j_1, \dots, j_n$.
2. Otherwise, choose from S a constraint

$$F : m \mid aj_i + b$$

such that b is an integer. Apply Theorem 7.15 to F .

3. If F is unsatisfiable, return.
4. Otherwise, instantiate the remaining constraints of S with each solution to j_i within the range $[1, \delta_i]$, and recursively solve.

Example 7.16. Consider the divisibility constraints of Example 7.12:

$$13 \mid -j - 26 \wedge 65 \mid 39 + j + k .$$

The system is already in triangular form for the variable order $k \prec j$. To solve the first constraint

$$\begin{array}{l} m \mid aj + b \\ 13 \mid -j + -26 \end{array} \quad \text{compute} \quad \begin{array}{l} d, p, q = \gcd(a, m) \\ 1, -1, 0 = \gcd(-1, 13) . \end{array}$$

Does $1 \mid -26$? Yes, so the solutions are given by

$$j = -(-1)\frac{-26}{1} + \lambda\frac{13}{1} = -26 + 13\lambda \quad \text{for } \lambda \in \mathbb{Z} .$$

Since $j \in [1, 13]$, only $\lambda = 3$ is relevant, providing the solution $j = 13$.

After substituting $j = 13$ into the second constraint, to solve

$$\begin{array}{l} m \mid ak + b \\ 65 \mid k + 52 \end{array} \quad \text{compute} \quad \begin{array}{l} d, p, q = \gcd(a, m) \\ 1, 1, 0 = \gcd(1, 65) . \end{array}$$

Does $1 \mid 52$? Yes, so the solutions are given by

$$k = -(1)\frac{52}{1} + \lambda\frac{65}{1} = -52 + 65\lambda \quad \text{for } \lambda \in \mathbb{Z} .$$

Since $k \in [1, 65]$, only $\lambda = 1$ is relevant, providing the solution $k = 13$. $j = 13$, $k = 13$ is the only solution to the divisibility constraints $13 \mid -j - 26 \wedge 65 \mid 39 + j + k$.

However, the additional conjunct

$$(k < -66j - 1794)\{j \mapsto 13, k \mapsto 13\}$$

is not true, so the formula of Example 7.12 is $\widehat{T}_{\mathbb{Z}}$ -equivalent to $\perp$ and is thus $\widehat{T}_{\mathbb{Z}}$ -unsatisfiable.

Alternatively, we could have enumerated all $13 \times 65 = 845$ possible disjuncts to discover that none of them simplifies to $\top$. In 844 of these disjuncts, at least one of the divisibility constraints is $\perp$.

Solve the triangulated system of Example 7.14 with variable order $j \prec k$, and verify that the solution is the same. The variable order used for triangulating the constraint system does not affect the solutions. ■

Solving divisibility constraints significantly improves the performance of QE on purely existential and purely universal formulae. What about formulae

with free variables or some quantifier alternation? For example, consider the formula

$$G : \forall y_1. \dots \forall y_m. \exists x_1. \dots \exists x_n. F[x_1, \dots, x_n, y_1, \dots, y_m] .$$

Eliminating the inner block produces the $\widehat{T_{\mathbb{Z}}}$ -equivalent formula

$$G' : \forall y_1. \dots \forall y_m. \bigvee_{j_1=1}^{\delta_1} \dots \bigvee_{j_n=1}^{\delta_n} F'[j_1, \dots, j_n, y_1, \dots, y_m] .$$

Order the free variables $\text{free}(F') = \{j_1, \dots, j_n, y_1, \dots, y_m\}$ so that the y_i 's precede the j_i 's. In the resulting triangulated system, as few constraints as possible contain y_i variables. Drop all resulting constraints that contain a y_i variable, and solve the remaining constraints. A variable j_i that does not appear in the final set of constraints must be instantiated to all values in its range $[1, \delta_i]$.

7.3 Quantifier Elimination over Rationals

QE for the theory of rationals $T_{\mathbb{Q}}$ is simpler than for $\widehat{T_{\mathbb{Z}}}$. Recall that $T_{\mathbb{Q}}$ has the following signature:

$$\Sigma_{\mathbb{Q}} : \{0, 1, +, -, =, \geq\} ,$$

where

- 0 and 1 are constants;
- + is a binary function;
- - is a unary function;
- and = and $\geq$ are binary predicates.

To be consistent with our presentation of Cooper's method, we switch from weak inequality $\geq$ to strict inequality $>$. Of course, they are interchangeable:

$$x \geq y \Leftrightarrow x > y \vee x = y \quad \text{and} \quad x > y \Leftrightarrow x \geq y \wedge \neg(x = y) .$$

7.3.1 Ferrante and Rackoff's Method

Given a $\Sigma_{\mathbb{Q}}$ -formula $\exists x. F[x]$ as input, where F is quantifier-free, the algorithm proceeds according to the following steps.

Step 1

Put $F[x]$ in NNF. The output $\exists x. F_1[x]$ is $T_{\mathbb{Q}}$ -equivalent to $\exists x. F[x]$ and is such that F_1 is a positive Boolean combination (only $\wedge$ and $\vee$) of literals.

Step 2

Replace literals according to the following $T_{\mathbb{Q}}$ -equivalences, applied from left to right:

$$\begin{aligned}\neg(s < t) &\Leftrightarrow t < s \vee t = s \\ \neg(s = t) &\Leftrightarrow t < s \vee t > s\end{aligned}$$

The output $\exists x. F_2[x]$ is $T_{\mathbb{Q}}$ -equivalent to $\exists x. F[x]$ and does not contain any negations.

Step 3

Solve for x in each atom of $F_2[x]$: for example, replace the atom

$$t < cx ,$$

where $c \in \mathbb{Z} \setminus \{0\}$ and t is a term not containing x , with

$$\frac{t}{c} < x .$$

Atoms in the output $\exists x. F_3[x]$ now have the form

- (A) $x < a$
- (B) $b < x$
- (C) $x = c$

where a, b, c are terms that do not contain x . $\exists x. F_3[x]$ is $T_{\mathbb{Q}}$ -equivalent to $\exists x. F[x]$.

Step 4

Construct the **left infinite projection** $F_{-\infty}$ from $F_3[x]$ by replacing

- (A) atoms $x < a$ by $\top$,
- (B) atoms $b < x$ by $\perp$,

and

- (C) atoms $x = c$ by $\perp$.

Construct the **right infinite projection** $F_{+\infty}$ from $F_3[x]$ by replacing

- (A) atoms $x < a$ by $\perp$,
- (B) atoms $b < x$ by $\top$,

and

- (C) atoms $x = c$ by $\perp$.

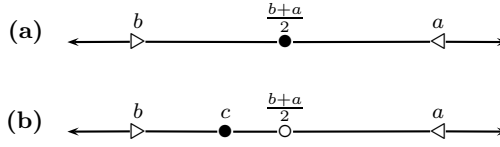


Fig. 7.2. Satisfying points: (a) $\frac{b+a}{2}$ (b) $\frac{c+c}{2}$

The left (right) infinite projection captures the case when small (large) $n \in \mathbb{Q}$ satisfy $F_3[n]$.

Let S be the set of a , b , and c terms from the **(A)**, **(B)**, and **(C)** atoms. Construct the final output

$$F_4 : F_{-\infty} \vee F_{+\infty} \vee \bigvee_{s,t \in S} F_3 \left[\frac{s+t}{2} \right],$$

which is $T_{\mathbb{Q}}$ -equivalent to $\exists x. F[x]$.

The disjunct $F_{-\infty}$ captures the possibility that all rationals n less than some value a^* satisfy $F_4[n]$. The disjunct $F_{+\infty}$ captures the symmetric case for large n . Finally, the last disjunct can be seen as capturing two possibilities. First, disjuncts in which s and t are the same terms check whether any term $s \in S$ satisfies $F_4[s]$. Second, consider the remaining $O(|S|^2)$ disjuncts in which s and t are different terms. In any $T_{\mathbb{Q}}$ -interpretation, $|S| - 1$ pairs $s, t \in S$ are adjacent; for such a pair, (s, t) is an interval in which no other $s' \in S$ lies. If $F_4[\frac{s+t}{2}]$, then it can be shown that every other point $n \in (s, t)$ also satisfies $F_4[n]$. In other words, $\frac{s+t}{2}$ represents the whole interval (s, t) . Since no single $T_{\mathbb{Q}}$ -interpretation is fixed in advance, all $O(|S|^2)$ pairs are considered (but see Exercise 7.8 for an optimization).

Figure 7.2 illustrates two cases in which a , b , and c terms are constant. Figure 7.2(a) visualizes the formula $b < x \wedge x < a$, for which $\frac{b+a}{2}$ is a satisfying point. Triangles represent inequalities; the solid circle represents $\frac{b+a}{2}$. All points in the interval (b, a) are satisfying, but $\frac{b+a}{2}$ is the representative of this interval. Figure 7.2(b) includes an additional literal, $x = c$ (the solid circle); now, $\frac{b+a}{2}$ (the open circle) is not a satisfying point, but $\frac{c+c}{2} = c$ is.

Example 7.17. Consider the $\Sigma_{\mathbb{Q}}$ -formula

$$\exists x. \underbrace{2x = y}_{F[x]}.$$

In Step 3, solving for x produces

$$F' : \exists x. x = \frac{y}{2}$$

so that $S = \{\frac{y}{2}\}$. The left $F_{-\infty}$ and right $F_{+\infty}$ infinite projections are both $\perp$, as F' contains a single **(C)** atom. Hence, simplifying

$$F_4 : \bigvee_{s,t \in S} \left(\frac{s+t}{2} = \frac{y}{2} \right)$$

reveals the $T_{\mathbb{Q}}$ -equivalent quantifier-free formula $\frac{y}{2} = \frac{y}{2}$, or $\top$. Therefore, $\exists x. F[x]$ is $T_{\mathbb{Q}}$ -valid. ■

Example 7.18. Consider the $\Sigma_{\mathbb{Q}}$ -formula

$$\exists x. \underbrace{3x + 1 < 10 \wedge 7x - 6 > 7}_{F[x]} .$$

Solving for x gives

$$F' : \exists x. \underbrace{x < 3 \wedge x > \frac{13}{7}}_{F_3[x]}$$

and $S = \{3, \frac{13}{7}\}$. Since $x < 3$ is an **(A)** atom and $x > \frac{13}{7}$ is a **(B)** atom, both $F_{-\infty}$ and $F_{+\infty}$ simplify to $\perp$, leaving

$$F_4 : \bigvee_{s,t \in S} \left(\frac{s+t}{2} < 3 \wedge \frac{s+t}{2} > \frac{13}{7} \right) .$$

$\frac{s+t}{2}$ takes on three expressions: $3, \frac{13}{7}$, and $\frac{\frac{13}{7}+3}{2}$. The first two expressions arise when s and t are the same terms. $F_3[3]$ and $F_3[\frac{13}{7}]$ both simplify to $\perp$ since the inequalities are strict; however,

$$F_3 \left[\frac{\frac{13}{7}+3}{2} \right] : \frac{\frac{13}{7}+3}{2} < 3 \wedge \frac{\frac{13}{7}+3}{2} > \frac{13}{7}$$

simplifies to $\top$. Thus, $F_4 : \top$ is $T_{\mathbb{Q}}$ -equivalent to $\exists x. F[x]$, so $\exists x. F[x]$ is $T_{\mathbb{Q}}$ -valid. ■

Example 7.19. Consider the $\Sigma_{\mathbb{Q}}$ -formula

$$G : \forall x. x < y .$$

To eliminate x , consider the subformula F of

$$G' : \neg(\underbrace{\exists x. \neg(x < y)}_{F[x]}) .$$

Step 2 rewrites F as

$$\exists x. y < x \vee y = x .$$

The literals are already in solved form for x in Step 3. Then

$$F_{-\infty} : \perp \vee \perp \quad \text{and} \quad F_{+\infty} : \top \vee \perp$$

simplify to $\perp$ and $\top$, respectively. Since $F_{+\infty}$ is $\top$, we need not consider the rest of Step 4, but instead declare that $\exists x. F[x]$ is $T_{\mathbb{Q}}$ -equivalent to $F_4 : \top$. Then G' is $\neg\top$, so that G is $T_{\mathbb{Q}}$ -equivalent to $\perp$. ■

Theorem 7.20 (Correct). *Given $\Sigma_{\mathbb{Q}}$ -formula $\exists x. F[x]$ in which F is quantifier-free, Ferrante and Rackoff's method returns a $T_{\mathbb{Q}}$ -equivalent quantifier-free formula.*

Exercise 7.9 asks the reader to prove the theorem.

A limited form of the block elimination optimization discussed in Section 7.2.4 can be adapted to this QE procedure: commute disjunction and existential quantification. This step reduces the size of the term set S in each subproblem.

7.4 ★Complexity

Fischer and Rabin proved the following lower bounds. The length n of a formula is the number of symbols.

Theorem 7.21 ($\widehat{T}_{\mathbb{Z}}$ Lower Bound). *There is a fixed constant $c > 0$ such that for all sufficiently large n , there is a $\widehat{\Sigma}_{\mathbb{Z}}$ -formula of length n that requires at least $2^{2^{cn}}$ steps to decide its validity.*

Theorem 7.22 ($T_{\mathbb{Q}}$ Lower Bound). *There is a fixed constant $c > 0$ such that for all sufficiently large n , there is a $\Sigma_{\mathbb{Q}}$ -formula of length n that requires at least $2^{2^{cn}}$ steps to decide its validity.*

Oppen analyzed Cooper's method to prove the following upper bound.

Theorem 7.23 ($\widehat{T}_{\mathbb{Z}}$ Upper Bound). *On a $\widehat{\Sigma}_{\mathbb{Z}}$ -formula of length n , Cooper's method requires deterministic time $2^{2^{pn}}$ for some fixed constant $p > 0$.*

Ferrante and Rackoff proved the following upper bound.

Theorem 7.24 ($T_{\mathbb{Q}}$ Upper Bound). *On a $\Sigma_{\mathbb{Q}}$ -formula of length n , Ferrante and Rackoff's method requires deterministic time $2^{2^{pn}}$ for some fixed constant $p > 0$.*

Closing the gap between the lower and upper bounds would require answering long-standing open questions in complexity theory.

7.5 Summary

Quantifier elimination is a standard technique for reasoning about theories in which satisfiability is decidable even with arbitrary quantification. This chapter presents the technique in the context of arithmetic over integers and over rationals or reals. It covers:

- *Quantifier elimination* in general. Based on structural induction, one only needs to consider the special case of formulae of the form $\exists x. F[x]$, in which F is quantifier-free but may contain free variables in addition to x ; arbitrary formulae may then be treated compositionally.
- *Elimination over integers, $T_{\mathbb{Z}}$* . The basic theory of integers does not admit quantifier elimination; it must be augmented with divisibility predicates. This situation, in which additional predicates are required to develop a quantifier elimination procedure, is common. The main idea of the procedure is to identify intervals with periodic behavior induced by the divisibility predicates.
- *Elimination over rationals, $T_{\mathbb{Q}}$* . The main idea of the procedure is to partition the rationals into a finite number of points and intervals.

The optimizations of Cooper's method, particularly solving divides constraints, make the procedure acceptably fast in practice on quantifier-free $\Sigma_{\mathbb{Z}}$ -formulae. However, faster decision procedures exist for deciding $\Sigma_{\mathbb{Q}}$ -satisfiability of quantifier-free $\Sigma_{\mathbb{Q}}$ -formulae; we study one in Chapter 8.

In addition to handling quantifiers, the algorithms of this chapter treat arbitrary Boolean combinations of literals. The decision procedures of subsequent chapters require the Boolean structure to be simple: formulae are just conjunctions of literals. Treating formulae with arbitrary Boolean structure directly avoids the potential exponential increase in size associated with converting to DNF.

Bibliographic Remarks

Presburger proves that arithmetic over the natural numbers without multiplication $T_{\mathbb{N}}$ is decidable [73]. Cooper presents the version of the quantifier-elimination procedure for $T_{\mathbb{Z}}$ that we describe [19]. Fischer and Rabin provide the lower bound on the complexity of the decision problem for $T_{\mathbb{Z}}$ [33], while Oppen analyzes Cooper's procedure to obtain an upper bound [68].

Ferrante and Rackoff describe the quantifier-elimination procedure that we present and the lower and upper complexity bounds on the problem [32].

Exercises

7.1 ($T_{\mathbb{Z}}$ does not admit QE). Prove Lemma 7.4. *Hint:* Apply structural induction; the base cases involve comparisons between ay and c , for constants a and c .

7.2 (QE for $\widehat{T_{\mathbb{Z}}}$). Apply quantifier-elimination to the following $\Sigma_{\mathbb{Z}}$ -formulae.

- $\forall y. 3 < x + 2y \vee 2x + y < 3$
- $\exists y. 3 < x + 2y \vee 2x + y < 3$

- (c) $\exists y. x = 2y \wedge y < x$
 (d) $\forall x. (\exists y. x = 2y) \rightarrow (\exists y. 3x = 2y)$

7.3 (QE for $\widehat{T_{\mathbb{Z}}}$). Construct new $\Sigma_{\mathbb{Z}}$ -formulae such that

- (a) the $F_{-\infty}$ component of the elimination simplifies to $\top$;
 (b) the $F_{-\infty}$ component of the elimination simplifies to $\perp$;
 (c) using the right infinite projection is better than using the left infinite projections.

In each case, describe the elimination.

7.4 (Block elimination in $T_{\mathbb{Z}}$). Apply quantifier-elimination to the following $\Sigma_{\mathbb{Z}}$ -formulae. Use block elimination.

- (a) $\exists x. \exists y. 2x + 3y = 7 \wedge x < y$
 (b) $\exists x. \exists y. 2x + 3y = 7 \wedge y < x$
 (c) $\exists x. \exists y. 2x + 3y = 7 \wedge x < y \wedge 0 < x \wedge 0 < y$
 (d) $\exists x. \exists y. 3x + 3y < 8 \wedge 8 < 3x + 2y$
 (e) $\exists x. \exists y. x = 2y \wedge \exists z. x = 3z$

7.5 (Divides constraints). Apply the divides-constraints elimination to the $\Sigma_{\mathbb{Z}}$ -formulae of Exercise 7.4.

7.6 (QE for $T_{\mathbb{Q}}$). Apply quantifier-elimination to the formulae of Exercise 7.2, but treat them as $\Sigma_{\mathbb{Q}}$ -formula.

7.7 (QE for $T_{\mathbb{Q}}$). Construct new $\Sigma_{\mathbb{Q}}$ -formulae such that

- (a) the $F_{-\infty}$ and $F_{+\infty}$ components of the elimination simplify to $\top$ and $\perp$, respectively;
 (b) the $F_{-\infty}$ and $F_{+\infty}$ components of the elimination simplify to $\perp$ and $\top$, respectively.

In each case, describe the elimination.

7.8 (Sufficient set). Step 4 of Ferrante and Rackoff's method examines terms $\frac{s+t}{2}$ for all $s, t \in S$, where S is the set of all a , b , and c terms. Describe a smaller set of terms that is still sufficient. According to this new definition, which terms should be examined in Example 7.18 and Exercise 7.6?

7.9 (★Theorem 7.20). Prove Theorem 7.20. *Hint:* Apply the strategy employed in the proof of Theorem 7.10.

7.10 (★Optimization problem). Consider the **optimization problem** $\max\{f(\bar{x}) : F[\bar{x}]\}$ in which the **objective function** $f(\bar{x})$ is a linear expression over the problem variables $\bar{x}$, and the **constraint** $F[\bar{x}]$ is a $\Sigma_{\mathbb{Z}}$ -formula such that $\text{free}(F[\bar{x}]) = \bar{x}$. The solution to the problem is the largest number n such that there exists some evaluation $\bar{v}$ of $\bar{x}$ for which $f(\bar{v}) = n$ and $F[\bar{v}]$ is true. Show how to use QE for $T_{\mathbb{Z}}$ to solve this optimization problem.